Powers of Ten

Where every place is ten of the one beside it.

THE BIG QUESTION

When you multiply by ten, why do the digits move instead of the decimal point?

HANDS-ON PROJECT

Zoom Atlas

Pick one real object and draw it at ten times, a hundred times and a thousandth of its size, labelling every jump.

FLUENCY GAME

Decimal Duel

Draw four digit cards, build the largest number you can, and defend it against your opponent's.

AXIOM · MISSION COMMANDER, NINE TOURS

"Every place is ten of the one beside it. Learn that once and half of fifth grade stops being new material."

NAME ______________

DATE ______________

Pre-Assessment

Twelve items, six strands. Anything you already own, we skip.

This is not a test. Nothing here counts for a grade. Two items ask you to explain rather than calculate — write the sentence anyway, even if it feels obvious.

STRAND A
PLACE VALUE

A1 The 4 in 400 is worth how many times the 4 in 40?

A2 One tenth of 500?

STRAND B
DECIMALS

B1 Write four hundred six thousandths as a decimal.

B2 What does the 5 in 0.05 mean?

STRAND C
× AND ÷ BY TEN

C1 $3.5 \times 100 = ?$

C2 $42 \div 100 = ?$

STRAND D
EXPONENTS

D1 What number is 10^3?

D2 How many zeros does 10^5 have?

STRAND E
COMPARING

E1 Which is larger, 0.406 or 0.41?

E2 Is 0.7 the same as 0.70? Say why.

STRAND F
ROUNDING

F1 Round 4.7 to the nearest whole number.

F2 Round 2.451 to the nearest hundredth.

TEACHER PAGE · DO NOT HAND TO STUDENT

Skip Chart

Score the pre-assessment, then cross days off the four-week calendar.

2 / 2 → SKIP
1 / 2 → COMPRESS
0 / 2 → TEACH

STRAND	ITEMS	COVERS	SCORE	IF 2/2, REMOVE
A · Place value	A1 A2	Week 1 · lesson 1.1	___ / 2	Day 1.1. Warm-Up tier drops to 3 items all mission.
B · Decimals	B1 B2	Week 1 · lessons 1.2 and 1.3	___ / 2	The Warm-Up halves of 1.2 and 1.3. Keep the Core — thousandths are new this year.
C · × and ÷ by ten	C1 C2	Week 2 · lessons 2.1 and 2.2	___ / 2	Compress Week 2 to three days and spend the extra two on the Zoom Atlas.
D · Exponents	D1 D2	Week 2 · lesson 2.3	___ / 2	Day 2.3. Exponent notation still appears in every later mission, so keep it visible.
E · Comparing	E1 E2	Week 1 · lesson 1.4	___ / 2	Nothing. If E2 is explained well, move 1.4 to Challenge only.
F · Rounding	F1 F2	Week 3 · lessons 3.1 and 3.2	___ / 2	Days 3.1 and 3.2. Keep 3.3 — choosing the place is the part that matters.

ONE NOTE BEFORE YOU START

This mission underwrites the whole year. The multiplication algorithm, long division and every decimal operation all lean on “each place is ten of the one to its right.” If a strand looks shaky, spend the extra days here rather than borrowing them back later.

MISSION 01 · WEEK 1 · MONDAY · LESSON 1.1

Ten Times Bigger

NAME ___________

DATE ___________

Each place is ten of the place to its right. Say it while you write it.

Warm-Up

WHAT EACH DIGIT IS WORTH

the 4 in 40	the 4 in 400	ten times 30	ten times 700	a tenth of 500	a tenth of 60
____	____	____	____	____	____

THE PLACE-VALUE LADDER · EVERY RUNG IS TEN OF THE ONE BELOW

1000	100	10	1	0.1	0.01
thousands	hundreds	tens	ones	tenths	hundredths

Going left, each rung is ten times the one before. Going right, each is one tenth. The ladder does not stop at the ones column — it just keeps going, and the decimal point marks where the whole numbers end.

Core

1. The 7 in 7,000 is how many times the 7 in 700? ....
2. The 3 in 300 is how many times the 3 in 3? ....
3. One hundred times 40 ....
4. One hundredth of 900 ....
5. The 5 in 50,000 is worth how much? ....

Challenge

NOT GRADED

C1. The 6 in 60,000 is how many times the 6 in 60? Explain the jump in one sentence.

....

C2. A digit moves two places left. Its value is multiplied by what? What if it moves three places right?

....

AXIOM: Say the place name out loud as you write each digit. Every mistake in this mission comes from a digit sitting in the wrong column.

NAME ___________

DATE ___________

Down to Thousandths

The same rule running the other way. Three places past the point.

Warm-Up

WRITE IT AS A DECIMAL

four tenths	seven hundredths	three thousandths	25 hundredths	one half	one quarter

READING 0.406 OUT LOUD

ONES	.	TENTHS	HUNDREDTHS	THOUSANDTHS
0	.	4	0	6

Say "four hundred six thousandths" — not "zero point four zero six." The last place gives the whole number its name, and saying it properly is what makes comparing decimals easy on Thursday.

Core

1. Write four hundred six thousandths ....

2. Write two and thirty-five hundredths ....

3. Write 0.60 more simply ....

4. Write sixty-two thousandths ....

5. How many hundredths are in 0.3? Draw or explain how you know.

....

Challenge

C1. Write nine and nine thousandths. Then say why the zeros matter.

....

C2. How many thousandths are in 0.25? Show the reasoning, not just the answer.

....

NAME ___________

DATE ___________

Read It Out Loud

The word for the last place is the word for the whole decimal.

Warm-Up NAME THE LAST PLACE

0.5	0.05	0.005	0.25	1.7	3.041

THREE WAYS TO WRITE THE SAME NUMBER

STANDARD	WORD FORM	EXPANDED
2.35	two and thirty-five hundredths	2 + 0.3 + 0.05

Core CONVERT BETWEEN THE THREE FORMS

1. 4.06 in words ....
2. “Seven and eight tenths” as a decimal ....
3. 0.512 in expanded form ....
4. 3 + 0.4 + 0.007 as one decimal ....
5. Why is 0.4 the same as 0.40? Write one sentence. ....

Challenge

C1. Write a decimal that has a 7 in the thousandths place and a 4 in the tenths place, and nothing else.

....

C2. Somebody reads 0.5 as “zero point five.” Somebody else says “five tenths.” Which one helps you compare it to 0.45, and why?

....

NAME ___________

DATE ___________

Compare and Order

Line up the points, not the digits. Trailing zeros are free.

Warm-Up

CIRCLE THE LARGER

0.5 or 0.35	0.7 or 0.70	0.9 or 0.89	0.06 or 0.6	1.2 or 1.19	0.4 or 0.400

WHY 0.406 IS SMALLER THAN 0.41

4	0	6
4	1	0

Both start with 4 tenths, so the tenths tell you nothing. Move one place right: 0 hundredths against 1 hundredth. That single column settles it — **0.41 is larger**, even though 406 looks bigger than 41.

Adding a trailing zero to 0.41 makes the comparison obvious and changes nothing about its value.

Core

1. Larger: 0.406 or 0.41 ....

2. Larger: 0.08 or 0.075 ....

3. Larger: 1.5 or 1.45 ....

4. Larger: 0.3 or 0.2999 ....

5. Put these in order, smallest first: 0.6 0.55 0.506 0.65

Challenge

C1. Write a decimal between 0.4 and 0.41. Then write another one between your answer and 0.41.

C2. Can you always find a decimal between two others? Try it with 0.999 and 1, then say what you think.

Decimal Duel

NAME ___________

DATE ___________

Four digit cards. Build the largest number you can, then defend it.

How to play: shuffle digit cards 0–9. Each player draws four and arranges them into a decimal of the form _ . _ _ _ — one whole, three decimal places. Larger number wins the round. Say your number out loud in words before comparing; mis-saying it loses the round.

Warm-Up

Digits 4, 7, 1, 9 — the biggest whole number you can build

The same digits — the smallest whole number

Play six rounds

ROUND	MY DIGITS	MY NUMBER	THEIR NUMBER	WON?
1				
2				
3				
4				
5				
6				

Challenge

C1. Which position matters most when you place your digits, and which matters least? Say why.

C2. Digits 2, 5, 8 — build the value closest to 1 in the form 0.___ . What did you have to think about?

Weeks 2–4

Every day below has five graded sets waiting in Practice Bay.

QUIZ · end of week 3
TEST · end of week 4

Week 2 · Multiply by Ten

IN THE APP

Multiplying by ten does not move the point. It moves every digit one place left, and the point stays exactly where it always was.

MON 2.1
× 10, 100, 1000
Watch the digits move. The point never does.

TUE 2.2
÷ 10, 100, 1000
The same journey in reverse.

WED 2.3
Exponent shorthand
10^3 means three tens multiplied, which is three places.

THU 2.4
Patterns in zeros
Why 4 × 100 ends in two zeros and 0.4 × 100 does not.

FRI
Zoom Atlas begins
Pick the object. Draw it at true size first.

Week 3 · Rounding

QUIZ FRIDAY

Rounding is a decision about which place matters, and this week he has to say which place he chose and why.

MON 3.1
Round to a named place
Find the place, look one to its right, decide.

TUE 3.2
Rounding decimals
Same method, smaller places.

WED 3.3
Which place matters
Money rounds to hundredths. Distance rarely does.

THU 3.4
Estimate to check
Round first, compute second, compare.

FRI
Mid-unit quiz
Page 10. Eight items, 85% to keep flying.

Week 4 · Proof

UNIT TEST

The Zoom Atlas gets finished and defended, and the Mission 01 test closes the mission.

MON 4.1
Finish the atlas
Every jump labelled with its power of ten.

TUE 4.2
Explain a jump
Pick one panel and say what moved and what stayed.

WED 4.3
Mixed review
Place value, powers and rounding in one set.

THU
Error journal sweep
Fix only what repeats.

FRI
Mission 01 test
Pages 11–12. Twelve items plus one explanation.

Mid-Unit Quiz

NAME ______________

SCORE ______ / 8

Eight items from Weeks 1 to 3. 85% to keep flying — that's 7 out of 8.

1. The 7 in 7,000 is how many times the 7 in 700? ....

2. Write four hundred six thousandths as a decimal. ....

3. 0.06×100 ....

4. $850 \div 10^2$ ....

5. Write 10^4 as an ordinary number. ....

6. Round 2.451 to the nearest hundredth. ....

7. Which is larger, 0.406 or 0.41? ....

8. A friend says "multiplying by ten moves the decimal point one place right." The answer they get is correct, but the sentence is wrong. Explain what actually moves.

....

....

Marking note: item 8 is worth the same as any other. Accept any answer saying the digits shift places while the point stays fixed — the wording matters less than the idea.

NAME ______________

DATE ______________

Unit Test · Part 1

Items 1–8. No time limit. Say each decimal out loud before you write anything.

1. The 3 in 300 is how many times the 3 in 3? ....

2. Write sixty-two thousandths as a decimal. ....

3. How many hundredths are in 0.3? ....

4. 2.4×10^3 ....

5. $7 \div 1000$ ....

6. Round 9.96 to the nearest tenth. ....

7. Round 0.0475 to the nearest thousandth. ....

8. Which power of ten takes 0.7 to 700? ....

CONTINUES ON PAGE 12

Unit Test · Part 2

SCORE ______ / 12

BAND ______________

Items 9–12 and the Big Question.

9. Put in order, smallest first: 0.6   0.55   0.506   0.65

....

10. 0.035×10^4

....

11. Write 3 + 0.4 + 0.007 as one decimal, then say it in words.

....

12. A model is one thousandth of true size. The real object is 4,000 mm long. How long is the model, and how do you know?

....

EXPLAIN YOUR THINKING · THE BIG QUESTION

When you multiply by ten, why do the digits move instead of the decimal point?

Use one number as your example and show it before and after. Scored on reasoning, not neatness.

....

....

....

....

....

11–12
Gold band. The Magnitude Badge is earned.

9–10
Silver band. One reteach day, then the badge.

≤ 8
Bronze. Reteach the weak strand and retest that part only.

TEACHER PAGE · ANSWER KEY

Key A

Pre-assessment, Lesson 1.1 and Lesson 1.2.

Mark the reasoning, not the handwriting. Any correct method counts.

Pre-assessment

A1 · 10 | A2 · 50 | B1 · 0.406 | B2 · five hundredths

C1 · 350 | C2 · 0.42 | D1 · 1000 | D2 · 5

E1 · 0.41 | E2 · yes — the trailing zero adds no value | F1 · 5 | F2 · 2.45

Lesson 1.1 · Ten Times Bigger

Warm-Up · 40 | 400 | 300 | 7000 | 50 | 6

Core · 1 · 10 | 2 · 100 | 3 · 4000 | 4 · 9 | 5 · 50,000

C1 · 1000, because it is three jumps left and each jump is ten times. C2 · two places left multiplies by 100; three places right divides by 1000.

Lesson 1.2 · Down to Thousandths

Warm-Up · 0.4 | 0.07 | 0.003 | 0.25 | 0.5 | 0.25

Core · 1 · 0.406 | 2 · 2.35 | 3 · 0.6 | 4 · 0.062 | 5 · 30 hundredths

C1 · 9.009 — the zero holds the hundredths place open, and without it the 9 would mean nine hundredths. C2 · 250 thousandths.

TEACHER PAGE · ANSWER KEY

Key B

Lesson 1.3, Lesson 1.4 and Friday's game.

Where a question asks for a sentence, no sentence means no mark — even with the number right.

Lesson 1.3 · Read It Out Loud

Warm-Up · tenths | hundredths | thousandths | hundredths | tenths | thousandths

Core · 1 · four and six hundredths | 2 · 7.8 | 3 · 0.5 + 0.01 + 0.002 | 4 · 3.407

Item 5 · because the extra zero sits in a place worth nothing — four tenths and forty hundredths are the same amount. C1 · 0.407. C2 · “five tenths” helps, because it names a place you can line up against the 4 tenths in 0.45.

Lesson 1.4 · Compare and Order

Warm-Up · 0.5 | equal | 0.9 | 0.6 | 1.2 | equal

Core · 1 · 0.41 | 2 · 0.08 | 3 · 1.5 | 4 · 0.3 | 5 · 0.506, 0.55, 0.6, 0.65

C1 · many answers, e.g. 0.405 then 0.4055. C2 · yes, always — you can keep adding decimal places. This is worth a proper conversation rather than a mark.

Friday · Decimal Duel

Warm-Up · 9741 | 1479 Rounds · his own; the table should show six completed rows

C1 · the ones place matters most and the thousandths least, because each place is ten times the one to its right. C2 · 0.852 — the largest digit has to go in the tenths place, and only then does the next one matter.

Key C

Mid-unit quiz, unit test, and the error journal sheet.

Quiz is out of 8, test out of 12. 85% is 7/8 and 11/12.

Mid-unit quiz

1 · 10

2 · 0.406

3 · 6

4 · 8.5

5 · 10,000

6 · 2.45

7 · 0.41

8 · the digits move, not the point

Unit test

1 · 100

2 · 0.062

3 · 30

4 · 2400

5 · 0.007

6 · 10

7 · 0.048

8 · 1000

10 · 350

11 · 3.407

Items 9, 12 and the Big Question

9 · 0.506, 0.55, 0.6, 0.65. **12** · 4 mm — dividing by a thousand moves every digit three places right. **Big Question** · full marks for any answer showing one number before and after, with the digits shifting one column left and the point staying put. An answer that says “the point moves right” with a correct calculation scores half: the arithmetic is fine, but the sentence teaches the wrong thing next year when they meet 0.4 × 100.

Error journal · Mission 01

ONE LINE PER MISTAKE · STUDENT PAGE

Write what went wrong, not what the right answer was. On Thursday of Week 4, read the whole list and fix only what appears twice.

DAY	WHAT I GOT WRONG, AND WHY